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CO₂ Flooding Optimization Using Artificial Neural Networks: Enhancing Oil Recovery and Carbon Sequestration

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Abstract

The rising global demand for hydrocarbons underscores the need to optimize oil recovery from existing reservoirs. Conventional methods, such as primary and secondary recovery, leave significant volumes of oil unrecovered, while Enhanced Oil Recovery (EOR) techniques—particularly CO₂ flooding—offer the dual benefit of increased oil recovery and carbon sequestration. CO₂ flooding improves oil displacement efficiency by reducing viscosity, with the Minimum Miscibility Pressure (MMP) being a critical parameter for achieving miscibility. However, accurate MMP determination remains challenging, as traditional experimental methods are costly and often limited in addressing impure CO₂ streams. This research addresses these limitations by developing an Artificial Neural Network (ANN) model trained on a comprehensive dataset encompassing both pure and impure CO₂ cases. The ANN demonstrated superior predictive performance compared to traditional methods, achieving an R² of 0.95 during training and 0.94 during testing. Key factors influencing MMP, including CO₂ purity, reservoir temperature, and oil composition, were systematically analyzed, revealing that higher temperatures and heavier hydrocarbons increase MMP, while intermediate hydrocarbons reduce it. CMG reservoir simulations further confirmed that injecting CO₂ at pressures above the predicted MMP results in the highest oil recovery and CO₂ retention, enhancing both EOR efficiency and carbon sequestration. To ensure practical field application, the ANN model was integrated into a mobile application, MMP Predictor, providing real-time, offline MMP predictions with minimal computational overhead. This AI-driven tool improves decision-making, reduces operational costs, and advances CO₂-EOR practices by providing a scalable, accurate, and field-ready method for MMP prediction that supports sustainable oil recovery and climate change mitigation efforts.

Keywords: Minimum Miscibility Pressure (MMP), CO₂ Enhanced Oil Recovery (EOR), Artificial Neural Networks (ANN), Carbon Sequestration, Sustainable Reservoir Management

Introduction

The global demand for hydrocarbons continues to rise, necessitating the optimization of oil recovery from existing reservoirs to meet energy needs sustainably (BP Statistical Review 2022). Conventional primary and secondary recovery methods often leave a significant portion of oil unrecovered, highlighting the critical role of Enhanced Oil Recovery (EOR) techniques (Green and Willhite, 2018). Among these, CO₂ flooding stands out for its dual benefits: enhancing oil recovery through miscible displacement and contributing to carbon sequestration, thereby mitigating greenhouse gas emissions (Alvarado and Manrique, 2010).

A key parameter in CO₂-EOR projects is the Minimum Miscible Pressure (MMP), which defines the pressure required for CO₂ and crude oil to achieve miscibility, maximizing displacement efficiency (Chen et al., 2020). Accurate MMP determination is essential for designing effective injection strategies, yet traditional methods—such as slim-tube experiments, empirical correlations, and equation-of-state (EOS) models—are often costly, time-consuming, or limited in handling complex reservoir conditions, particularly with impure CO₂ streams.

Recent advancements in artificial intelligence (AI), particularly Artificial Neural Networks (ANNs), offer a promising alternative for MMP prediction. ANNs can model nonlinear relationships between reservoir variables (e.g., temperature, oil composition, CO₂ purity) and MMP with high accuracy, outperforming conventional methods. However, existing ANN-based studies often rely on limited datasets or overlook the impact of impurities in CO₂ streams, which are common in industrial applications.

Traditional experimental methods like slim-tube tests provide accurate MMP measurements but are time-consuming and expensive (Rathmell et al., 1971). While empirical correlations and EOS models offer faster predictions, they often lack reliability for complex oil compositions and impure CO₂ streams (Shokir, 2007). To address these challenges, recent studies have increasingly adopted AI techniques, especially ANNs, for improved MMP prediction. Early ANN models showed better accuracy but were limited by small datasets and lacked thorough consideration of CO₂ impurities, which play a critical role in miscibility behavior (Yuan et al., 2004).

Dehghani (2008) introduced a Genetic Algorithm–Artificial Neural Network (GA-ANN) model using 46 data points, achieving an average absolute percentage error of 4.9%, outperforming several empirical methods. Shokrollahi (2013) proposed a Least-Squares Support Vector Machine (LSSVM) model for pure and impure CO₂ systems, yielding an R² of 0.903 and RMSE of 1.69 using 147 experimental values. Hemmati-Sarapardeh (2016) developed a feed-forward ANN with the Levenberg–Marquardt algorithm, achieving an R² of 0.872 and AAPRE of 10.46% on 147 MMP records, validating its reliability across CO₂ compositions. Karkevandi (2017) optimized an ANFIS model using five evolutionary algorithms and 270 MMP data points, with PSO achieving the best result—an average percent relative error of 7.53%. More recently, Al-Khafaji (2023) assessed five machine learning models, with KNN and SVR showing superior performance, especially in datasets with fewer samples. Gas composition and temperature were the most influential factors identified. Soliman (2024) evaluated ANN, Random Forest, KNN, and a Gradient Boosting Algorithm (GBA) with Grid Search on 205 MMP records; the GBA model achieved the highest accuracy with R² values of 0.999 (training) and 0.865 (validation), benefiting from its regression power and optimized hyperparameters.

This study addresses the identified gaps of small, fragmented datasets, limited consideration of CO₂ impurities, and insufficient model optimization by developing a robust ANN model trained on a comprehensive dataset encompassing both pure and impure CO₂ cases. The model's performance is validated against traditional methods and integrated with reservoir simulations using CMG software to evaluate its practical implications for oil recovery and CO₂ sequestration. Furthermore, the ANN model is deployed in a mobile application, *MMP Predictor*, enabling real-time, field-ready predictions with minimal computational overhead.

Building on this approach, the objectives of this research are threefold: first, to enhance the accuracy of MMP prediction in CO₂ flooding across diverse reservoir conditions; second, to investigate the influence of CO₂ purity on miscibility behavior and its implications for optimizing EOR and CO₂ sequestration; and third, to provide a practical, accessible prediction tool through mobile deployment, supporting efficient decision-making in the field. By integrating AI-driven modeling with real-world EOR operations, this work advances sustainable hydrocarbon recovery and contributes to climate change mitigation efforts.

Methodology

This section outlines the approach used to develop an AI-based model for MMP prediction in CO₂ flooding. It details data collection, statistical analysis, and preprocessing steps to ensure model accuracy and reliability. A reservoir model was built using CMG software to simulate CO₂ injection under various conditions, providing a comparative benchmark. The trained ANN model was then deployed in a mobile application for real-time MMP prediction, enhancing accessibility and usability for field applications.

Data Collection and Description

The dataset was compiled from [Alston et al. \(1985\)](#), [Rathmell et al. \(1971\)](#), [Shokir \(2007\)](#), [Yuan et al. \(2004\)](#), and [Yuan \(2003\)](#) forming a single unified dataset that includes both pure and impure CO₂ cases to ensure generalized applicability. The dataset encompasses cases of CO₂ injection, including both pure and mixed CO₂ compositions. This unified dataset allows for a comprehensive analysis of how the presence of additional components influences CO₂ flooding efficiency and MMP requirements. The dataset includes key parameters such as reservoir temperature, oil composition, the percentage of CO₂ in the injected gas, and the recorded MMP. These parameters serve as the foundation for developing an ANN model and analyzing CO₂ purity in EOR operations. The unified dataset provides a holistic view of CO₂ behavior under various conditions, enabling the model to learn from a broader range of scenarios. This approach ensures that the model can generalize well across different CO₂ compositions, including both pure and impure cases.

Statistical Analysis

An exploratory statistical analysis was conducted to examine relationships within the dataset and guide model development. Key patterns and trends were identified to improve predictive accuracy and assess the effects of CO₂ purity on MMP. Key reservoir and fluid properties, including temperature, intermediate component mole percent (C₂–C₆), heavier component molecular weight (C₇+), and MMP, were analyzed using descriptive statistics, as summarized in [Table 1](#).

Table 1—A statistical analysis of the input and output data used in this study.

	Temperature	C ₂ –C ₆ mol%	MW of C ₇	CO ₂ Purity	MMP
min	71.00	2.00	139.00	60.00	1100.00
max	300.00	43.50	319.70	100.00	5000.00
mean	179.67	23.26	200.64	88.01	2753.47
std	48.97	8.41	39.49	14.70	828.82
kurtosis	–0.46	–0.85	–1.02	–0.90	–0.58
skewness	0.37	0.28	–0.11	–0.79	0.23
mode	160.00	35.57	245.40	100.00	1879.00

The dataset presents temperature values spanning from 71°F up to 300°F, averaging around 179.67°F. Intermediate component concentrations (C₂–C₆ mole percent) vary significantly, ranging between 2.00% and 43.50%, with an average of 23.26%. The molecular weight for heavier hydrocarbon components (C₇+) is also analyzed.

shows a broad distribution from 139.00 to 319.70, having an average molecular weight of approximately 200.64. Additionally, CO₂ purity exhibits a wide range, from as low as 60% up to complete purity at 100%, averaging about 88.01%. The dataset also includes MMP values, ranging from 1100.00 psia to as high as 5000.00 psia, with an average value of 2753.47 psia.

Distribution patterns differ among these variables; temperature, C₂-C₆ mole percent, and MMP exhibit slight right-skewness, indicating a tendency toward lower values with fewer higher-value outliers. Conversely, C₇₊ molecular weight and CO₂ purity show slight left-skewness, suggesting a prevalence of higher values with relatively fewer lower-value observations.

Visualization techniques were employed to further understand the dataset's distribution and relationships, as shown in Fig. 1. The histogram for temperature, Fig. 1(a) shows a unimodal distribution with a peak around 160°F. The histogram for C₂-C₆ mole percent, illustrated in Fig. 1(b), reveals a slightly right-skewed distribution, with most values clustered around the mean. The histogram for C₇₊ molecular weight, demonstrated in Fig. 1(c) shows a relatively uniform spread of values. The histogram for MMP values, Fig. 1(d) is right-skewed, with most values clustered at the lower end of the range. While there is a concentration of data points around 90-100% purity in Fig. 1(e), a significant portion of the dataset includes CO₂ streams with lower purity levels (60-90%), suggesting that impure CO₂ streams are well-represented in the dataset. This distribution highlights the importance of considering both pure and impure CO₂ cases in MMP prediction models.

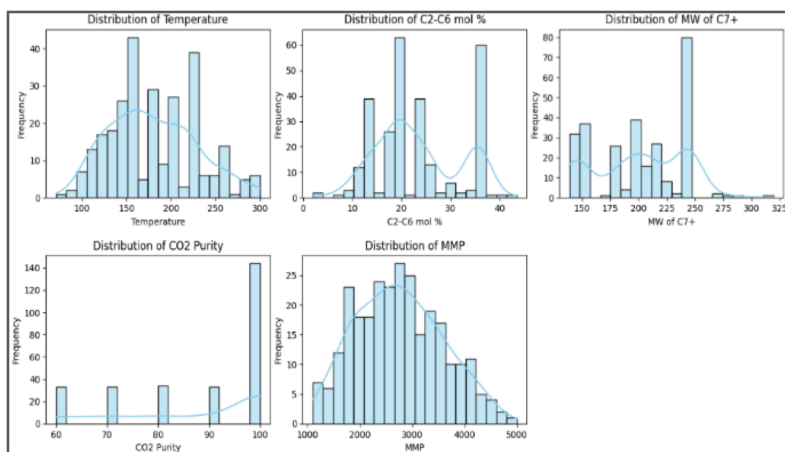


Figure 1—Distribution of key reservoir and fluid properties in the CO₂-dataset: (a) Temperature, (b) C₂-C₆ mole %, (c) C₇₊ molecular weight, (d) MMP, and (e) CO₂ purity.

The correlation heatmap in Fig. 2 provides a comprehensive correlation analysis for the dataset. Key observations include a strong positive correlation between reservoir temperature and MMP, a moderate negative correlation between the mole percent of intermediate hydrocarbons C₂-C₆ and MMP, a weaker positive correlation between the molecular weight of heavier components C₇₊ and MMP, and a negative correlation between CO₂ purity and MMP. These visualizations provide a clear understanding of the dataset's distribution and the relationships between key parameters, which are essential for developing an accurate ANN model.

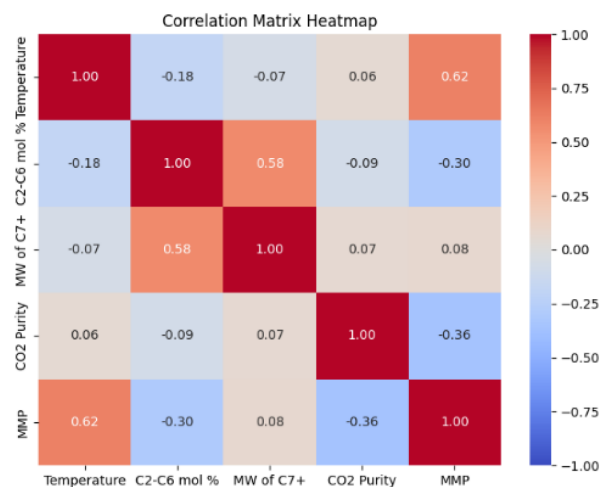


Figure 2—Correlation heatmap for the CO₂-dataset, showing the relationships between temperature, hydrocarbon composition, and MMP.

Moreover, the dataset underwent preprocessing to ensure quality, relevance, and suitability for modeling. This phase included data cleaning, normalization, and splitting for model training and evaluation. Outliers were managed using the Z-score method, while missing values were addressed through truncation to maintain data consistency. Standardization was applied to transform features, ensuring they have a mean of 0 and a standard deviation of 1.

To maintain predictive power, all available reservoir and fluid property variables were retained. The dataset was then divided into training (80%) and testing (20%) subsets, ensuring a balanced representation of pure and impure CO₂ cases for effective model training and evaluation.

Reservoir Simulation Using CMG

To evaluate the impact of CO₂ injection on oil recovery and carbon sequestration, a reservoir model was developed using CMG software. This section outlines the reservoir model setup, simulated scenarios, and the methodology for analyzing the results. The simulations were designed to assess the performance of CO₂ flooding under different injection pressures relative to the MMP, providing insights into optimal injection strategies for enhanced oil recovery and CO₂ storage.

A reservoir model was developed using CMG software to simulate typical conditions for EOR projects, as shown in Fig. 3. The model was developed based on the CMG CO₂ flooding and represents a multi-layered reservoir with a grid dimension of 50 × 50 × 10, covering a total reservoir area of 500 acres. The reservoir depth is 8000 ft, and the initial reservoir pressure is set at 3200 psi. A heterogeneous permeability distribution is implemented across the 10 layers, with a base permeability of 150 md to account for reservoir heterogeneity.

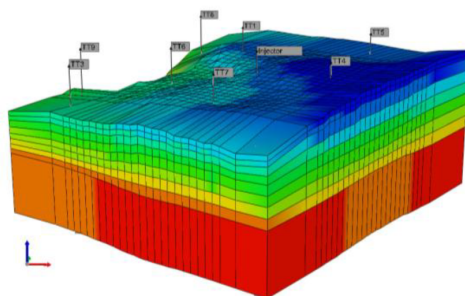


Figure 3—Reservoir model used in research

The MMP was calculated using CMG Winprop through the Multi Mixing-Cell method. The determined MMP is 2500 psi, based on values relevant to CO₂ flooding operations. The well properties such as injection and production pressures were configured based on conditions representative for CO₂ flooding.

The simulation work was conducted using CMG software, and multiple CO₂ injection scenarios were evaluated to assess their impact on oil recovery and CO₂ sequestration. The model was first run for 20 years to reduce the reservoir pressure below the MMP, enabling injection at sub-MMP conditions and allowing for testing of different injection effects. Three CO₂ injection scenarios were simulated to evaluate their effects on oil recovery and CO₂ sequestration. In the first scenario, CO₂ was injected at the Minimum Miscibility Pressure (MMP) of 2500 psi to assess optimal conditions for both recovery and storage. The second scenario involved injection at pressures below the MMP to examine the impact of immiscible flooding. In the third scenario, CO₂ was injected at pressures above 2500 psi to investigate whether elevated pressures enhance oil recovery and improve sequestration performance. For each scenario, key performance indicators such as oil recovery factor, CO₂ sequestration efficiency, and reservoir pressure profiles were monitored over the injection period. The simulations were conducted using vertical injectors and producers, maintaining consistent configurations across all scenarios to ensure comparability.

ANN Model Implementation in Android Application

The trained ANN model for MMP prediction was deployed in an Android application named MMP Predictor using Android Studio. The app allows users to input reservoir and fluid properties, perform unit conversions, and generate MMP predictions in real time. The model was converted into TensorFlow Lite (TFLite) format for efficient on-device inference, ensuring fast and offline predictions. The application workflow includes user input, automated unit conversion, data normalization, ANN inference, and result display. The integrated unit conversion module allows switching between field and SI units, ensuring compatibility with various datasets, and results are displayed with an option to view values in different unit formats.

Results and Discussion

Analysis of ANN Model Results

In this study, an ANN was developed to predict the MMP in CO₂ flooding. Various architectures and hyperparameters were tested to optimize model performance. Regularization techniques were implemented to mitigate overfitting, with L2 regularization proving most effective. The AdamW optimizer, with an adaptive learning rate of 0.001 and weight decay of 1E-04 was used to minimize the Mean Squared Error (MSE) loss function, chosen for its rigorous validation capabilities. A systematic sensitivity analysis was conducted to determine the optimal ANN architecture. Different numbers of hidden layers (1 to 3), neurons per layer (10 to 30), and activation functions (ReLU, LeakyReLU, Tanh, Sigmoid, etc.) were evaluated. The optimal configuration comprised one hidden layer with 20 neurons, employing ReLU activation functions and Batch Normalization to stabilize training. A linear activation function was used in the output layer for MMP prediction. The structure of the selected ANN is illustrated in Fig. 4, which shows the input layer with features such as reservoir temperature, C₆ mole percent, C₇₊ molecular weight, and CO₂ purity, the hidden layer with 20 neurons, and the output layer for MMP prediction. The model was trained on 80% of the dataset, with 20% reserved for testing.

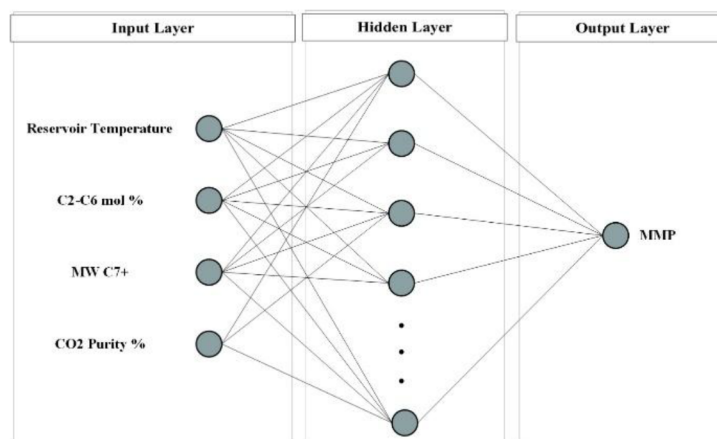


Figure 4—A schematic representation of the ANN architecture, detailing the input, hidden, and output layers.

The ANN model's performance was assessed using R^2 and MAPE. The optimal model achieved an R^2 of 0.95 during training and 0.94 during testing, along with a MAPE of 5.4% during training and 6.4% during testing, demonstrating strong predictive accuracy and reliability. The cross-plot of actual versus predicted MMP values in Fig. 5(a) for the training dataset and Fig. 5(b) for the testing dataset, demonstrates the model's accuracy, with most data points closely aligned along the 45-degree line. This strong correlation validates the ANN's ability to accurately predict MMP under varying reservoir conditions.

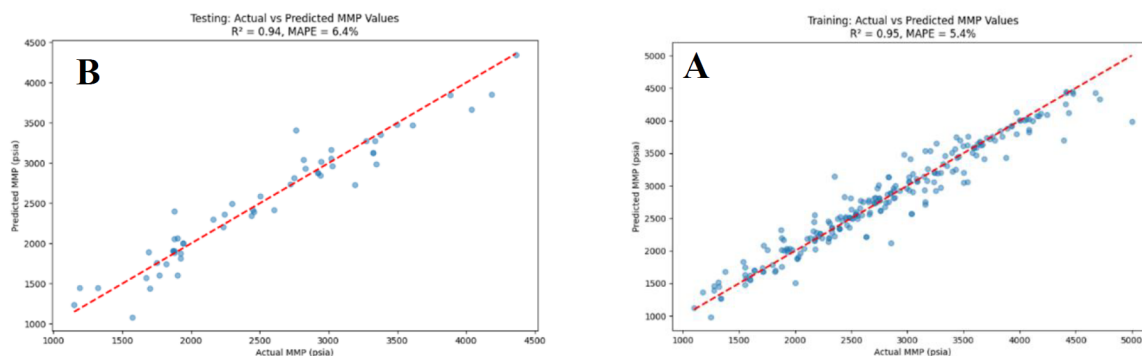


Figure 5—Regression plots of actual versus predicted MMP for (a) training dataset and (b) testing dataset.

Sensitivity Analysis of Key Parameters on MMP

A sensitivity analysis was conducted to evaluate the influence of CO₂ purity, reservoir temperature, and oil composition on MMP predictions. The trained ANN model was tested across a range of values for each parameter to assess their relative impact. The effects of CO₂ purity, reservoir temperature, and oil composition on MMP was studied.

CO₂ purity, represented as a percentage, was varied to analyze its effect on MMP. The results indicate a nonlinear relationship, where a decrease in CO₂ purity leads to an increase in MMP, reflecting reduced miscibility efficiency. Fig. 6 illustrates this trend, highlighting the sharp increase in MMP at lower purities.

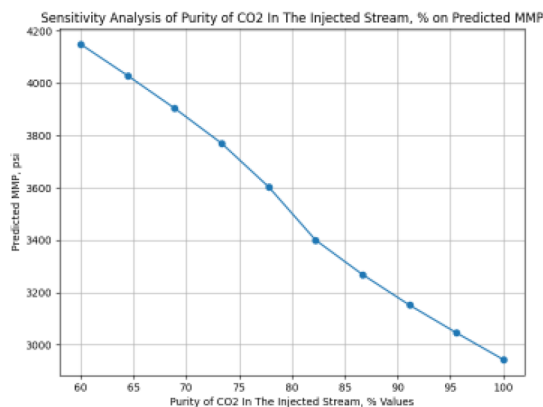


Figure 6—Effect of CO₂ purity in the injected stream on predicted MMP

Reservoir temperature was systematically varied from 70°F to 300°F to examine its impact. As shown in Fig. 7, MMP increases with temperature, suggesting that higher thermal energy reduces the effectiveness of CO₂ in achieving miscibility. This observation aligns with previous studies on CO₂ phase behavior in EOR applications.

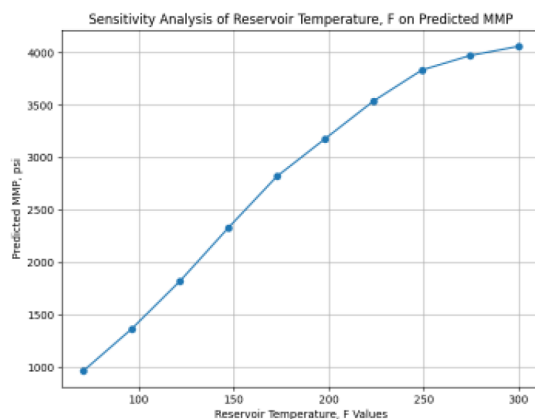


Figure 7—Effect of reservoir temperature on predicted MMP

Moreover, the oil composition was evaluated based on two key input features: intermediate components (C₂–C₆ mol percent) and heavier components (C₇₊ molecular weight). Higher concentrations of intermediate components generally result in lower MMP, indicating improved miscibility with CO₂. This trend is clearly observed in Fig. 8(a), where an increase in C₂–C₆ mol percent corresponds to a steady decline in MMP values. In contrast, heavier components exhibit the opposite effect; an increase in C₇₊ molecular weight correlates with higher MMP, as heavier fractions hinder miscibility. As shown in Fig. 8(b), reservoirs with higher C₇₊ molecular weight exhibit significantly elevated MMP values, emphasizing the role of heavy hydrocarbons in reducing miscibility.

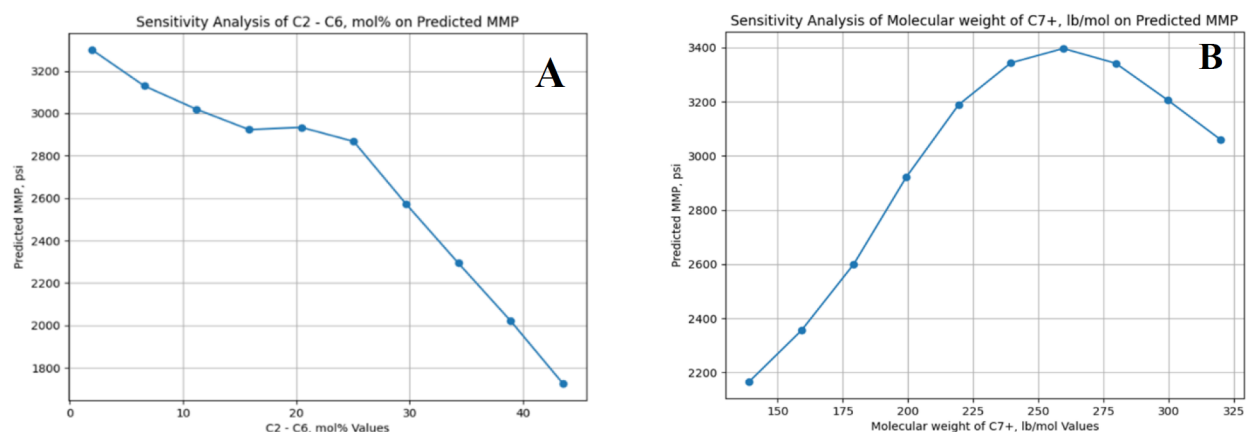


Figure 8—Effect of oil composition on predicted MMP: (A) Intermediate hydrocarbons (C_2 – C_6), (B) Heavy fractions (C_7).

Comparative Error Analysis

The performance of the ANN model was evaluated against traditional methods, including the correlations proposed by [Alston et al. \(1985\)](#), [Glaso \(1985\)](#), and [Yuan et al. \(2004\)](#), for predicting the MMP in both pure and impure CO_2 scenarios, see the Appendix. The comparison was based on key statistical metrics, including the Mean Absolute Percentage Error (MAPE) and the correlation coefficient (R). The results are summarized in [Table 2](#) and [Table 3](#).

Table 2—Statistical Performance Comparison for Pure CO_2 MMP Predictions

Method	MAPE (%)	R
Glaso (1985)	17.63	0.775
Alston et al. (1985)	29.49	0.748
Yuan et al. (2004)	28.02	0.778
ANN	7.28	0.957

Table 3—Statistical Performance Comparison for Impure CO_2 MMP Predictions

Method	MAPE (%)	R
Glaso (1985)	17.63	0.775
Alston et al. (1985)	29.49	0.748
Yuan et al. (2004)	28.02	0.778
ANN	7.28	0.957

As illustrated in [Table 3](#), the ANN model demonstrated superior accuracy with a MAPE of 7.28%, significantly lower than the MAPE values of the traditional methods, which ranged from 17.63% ([Glaso, 1985](#)) to 29.49% ([Alston et al., 1985](#)). Additionally, the ANN model outperformed the traditional methods in terms of correlation, with an R value of 0.957, compared to 0.748–0.778 for the other methods.

The ANN model demonstrated exceptional accuracy with a MAPE of 3.70%, significantly lower than the MAPE values of the traditional methods, which ranged from 21.39% ([Glaso, 1985](#)) to 42.05% ([Yuan et al., 2004](#)). The ANN model also outperformed the traditional methods in terms of correlation, with an R value of 0.978, compared to 0.409–0.527 for the other methods, as shown in [Table 3](#).

The ANN model consistently outperformed the traditional methods in both pure and impure CO_2 scenarios, achieving lower MAPE values and stronger correlation coefficients. The ANN model

demonstrated robustness across different CO₂ compositions, maintaining high accuracy and reliability even in impure CO₂ cases, where traditional methods struggled. The ANN model's ability to provide accurate and rapid MMP predictions makes it a valuable tool for optimizing CO₂ flooding operations, particularly in scenarios involving impure CO₂ streams.

In conclusion, the ANN model offers a significant improvement over traditional methods for MMP prediction, providing a more accurate, reliable, and scalable solution for CO₂ flooding optimization in both pure and impure CO₂ scenarios.

Analysis of CMG Simulation Results

The CMG simulations evaluated the impact of different CO₂ injection pressures on reservoir performance, focusing on oil recovery and CO₂ sequestration efficiency. The simulations compared three scenarios: injection above, at, and below the predicted MMP. The results highlight the critical role of accurate MMP prediction in optimizing CO₂ flooding operations.

The CMG simulations evaluated the impact of CO₂ injection pressures relative to the accurately predicted Minimum Miscible Pressure (MMP) on reservoir performance. Injections above MMP caused a rapid increase in reservoir pressure, peaking before gradually declining, demonstrating effective pressure maintenance. Injections at MMP stabilized pressure at intermediate levels, while injections below MMP led to a steady decline, highlighting the inefficiency of sub-optimal pressures. These findings, as illustrated in Fig. 9, underscore the importance of accurate MMP prediction for effective reservoir pressure management.

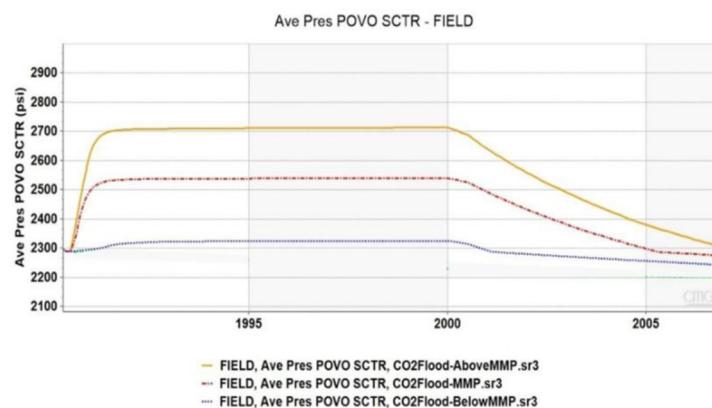


Figure 9—Average reservoir pressure (POVO SCTR) evolution in field for various CO₂ flooding strategies: base scenario (No injection), above MMP, at MMP, and below MMP.

In terms of oil recovery, the above MMP scenario initially recorded the highest oil production rates, but these rates declined over time. Injections at MMP provided stable and moderate recovery rates throughout the simulation, while the below MMP scenario consistently yielded lower oil recovery rates. The oil recovery factor was highest in the above MMP scenario, followed by at MMP and below MMP, as shown in Fig. 10, demonstrating that accurate MMP prediction is crucial for maximizing oil recovery in CO₂ flooding operations.

The sequestration potential of CO₂, determined by the difference between injected and produced volumes, was initially highest in the above MMP scenario. However, this scenario showed an increasing trend in CO₂ production over time, reducing its long-term sequestration efficiency. Injection at MMP resulted in moderate retention with stable production rates, while below MMP showed the lowest retention and sequestration potential. These results highlight the need to balance injection pressures to optimize both oil recovery and CO₂ sequestration.

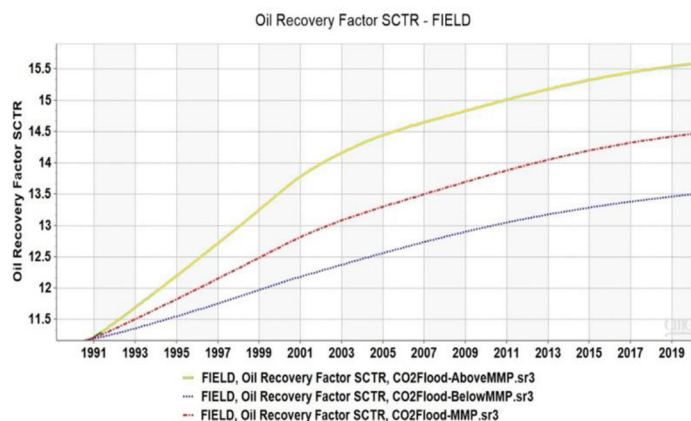


Figure 10—Comparison of Oil Recovery Factor over time for field under different CO₂ flooding scenarios: above MMP, at MMP, and below MMP.

Based on these results, it is recommended to inject CO₂ at pressures slightly above the accurately predicted MMP, as this approach optimizes CO₂ retention. However, the increasing CO₂ production trend in the above MMP scenario suggests that long-term sequestration efficiency may decline, requiring adjustments to injection strategies over time. The ANN model's ability to predict MMP with high accuracy is crucial for determining the optimal injection pressure and ensuring the success of CO₂ flooding operations while balancing recovery and sequestration goals.

Cost and Time Analysis

A comprehensive cost and time analysis were conducted to evaluate the efficiency of ANN-based MMP prediction compared to traditional methods such as the Slim Tube Test and CMG simulations. The analysis considered estimated average industry costs for setup, computational time, scalability, and long-term financial impact. These estimates provide a realistic comparison of the economic efficiency of ANN-based methods versus traditional approaches.

The Slim Tube Test was identified as both costly and time-consuming, with setup costs ranging from \$12,000 to \$35,000 and per-test costs between \$1,500 and \$5,000. Each test typically required 2 days to complete, making it impractical for rapid or large-scale predictions (Kantzas et al., 2012). CMG simulations, while more moderate in cost with setup expenses ranging from \$14,000 to \$65,000 and per-simulation costs of \$500 to \$1,000, also faced scalability limitations and required significant computational time, often taking 1 to 2 weeks to complete depending on the complexity of the simulation. In contrast, the ANN model required a higher initial investment, ranging from \$6,600 to \$18,500, but offered negligible per-prediction costs and near-instantaneous results.

This allowed for large-scale predictions to be conducted in a fraction of the time required by traditional methods. As illustrated in Fig. 11, the ANN model demonstrated superior cost and time efficiency across various scenarios. For a single prediction (Scenario 1), the ANN model was significantly more cost-effective and faster than both the Slim Tube Test and CMG simulations. This advantage became even more pronounced with 10 predictions (Scenario 2), and for 100 or more predictions (Scenario 3), the ANN model's scalability, low per-prediction cost, and rapid processing time made it the most efficient solution.

The ANN model reduced the time required for predictions by 96-99% compared to traditional methods. For instance, while the slim tube test could take 2 days and CMG simulations 1 to 2 weeks, the ANN model delivered results almost instantaneously. This dramatic reduction in time, combined with significant cost savings, underscores the ANN model's potential as a transformative tool for large-scale MMP predictions in CO₂ flooding projects. Its ability to provide rapid, accurate predictions at a lower long-term cost positions it as a highly efficient and scalable solution in the field.

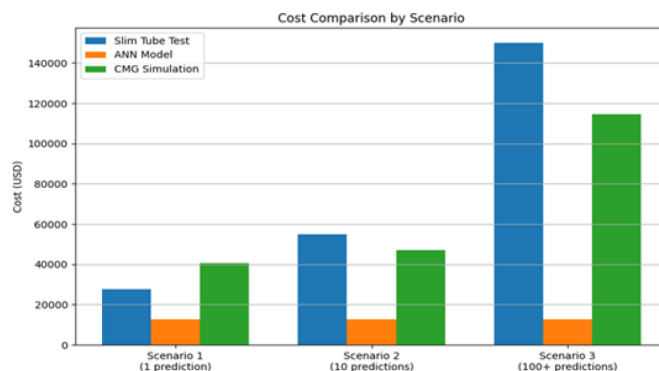


Figure 11—Cost efficiency of the ANN model compared to traditional methods (Slim tube test and CMG simulations) for 1, 10, and 100+ predictions, emphasizing the ANN model's scalability and cost savings.

Mobile Application Performance

The MMP Predictor app demonstrated high accuracy and reliability, providing real-time MMP predictions with minimal computational requirements. Predictions are processed within 200 ms, making the app highly efficient for field use. With a lightweight 25 MB size and offline functionality, the app is designed to be portable and accessible, offering reservoir engineers a practical tool for optimizing CO₂ flooding operations.

Fig. 12 illustrates the MMP Predictor interface, with Fig. 12 (a) showing the input screen where users enter parameters such as reservoir temperature (°F or °C), C₂–C₆ mol %, C₇₊ molecular weight (lb/mol), and CO₂ purity. Fig. 12 (b) displays the output screen, presenting the predicted MMP in MPa or psi (e.g., 17.18 MPa), with unit conversion options for both input and output.

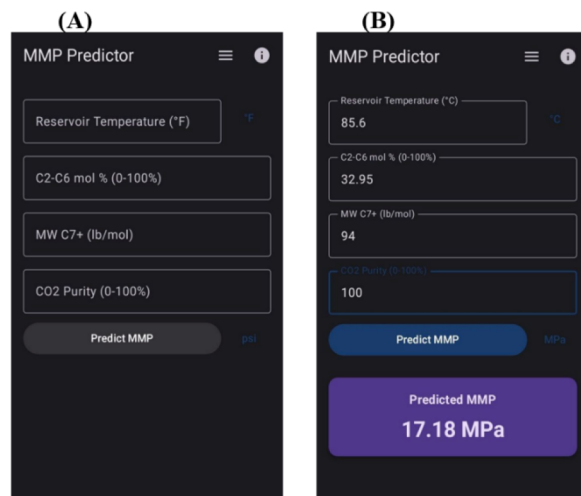


Figure 12—MMP predictor mobile application interfaces. (a) Input screen for reservoir parameters, (b) Output screen displaying predicted MMP (17.18 MPa).

The user-friendly design ensures seamless interaction, and the inclusion of unit conversion enhances usability for field engineers. Compared to CMG simulations, the app offers faster predictions, portability, and eliminates the need for extensive computational resources.

The MMP Predictor successfully integrates the ANN model into a mobile platform, providing fast, accurate, and accessible MMP predictions. Future improvements may include cloud-based storage and expanded parameter customization.

Conclusions

This research investigated CO₂ flooding optimization for enhanced oil recovery (EOR) using artificial neural networks (ANNs) to predict minimum miscible pressure (MMP), aiming to improve recovery efficiency and support carbon sequestration. An ANN model trained on data for pure and impure CO₂ showed high predictive accuracy with R² values of 0.95 (training) and 0.94 (testing), outperforming traditional correlations with MAPE values between 17.6% and 42.1%. The model revealed that CO₂ purity below 90% caused a nonlinear increase in MMP, requiring 9–15% higher pressures. Reservoir temperature showed a strong positive correlation, increasing MMP by ~238 psi per 100°F up to 300°F. Oil composition analysis showed a 10 mol% increase in C₂–C₆ reduced MMP by 12–18%, while heavier fractions (C₇₊ > 200 lb/mol) increased it by 22%. Moreover, CMG simulations confirmed that injection pressures slightly above predicted MMP provided the best balance between oil recovery and CO₂ sequestration, validating the model's practical utility. The ANN model also reduced costs by 96% compared to slim-tube tests, offering a faster and scalable alternative. Furthermore, the model was implemented in the "MMP Predictor" mobile app, delivering real-time results in under 200 milliseconds with only 25 MB of memory, enhancing field usability. Overall, this research offers an advanced, AI-driven approach to MMP prediction that improves decision-making, enhances EOR and sequestration efficiency, and significantly lowers operational costs.

Recommendations

Based on the findings of this study, future research should focus on expanding the training dataset, particularly for impure CO₂ conditions, to enhance the ANN model's accuracy and generalizability. Field trials are also recommended to validate the model's performance under real EOR conditions. Additionally, incorporating key reservoir characteristics such as permeability and porosity into the ANN model can improve predictions of CO₂ retention, leading to more efficient CO₂ flooding and sequestration.

Nomenclature

C ₂ –C ₆	= mole percent of intermediate hydrocarbon components, dimensionless, mol%
C ₇₊	= heavier hydrocarbon fraction, lbmol/mol, molecular weight [lb/mol]
MMP	= minimum miscibility pressure, m/Lt ² , psia [MPa]
P	= pressure, m/Lt ² , psi [kPa]
T	= temperature, Θ, °F [°C]
q	= injection or production rate, L ³ /t, STB/day [m ³ /day]
RF	= oil recovery factor, dimensionless, %
CO ₂ purity	= mole percent CO ₂ in injection stream, dimensionless, mol%
ANN	= artificial neural network (abbreviation; not included in symbols list)
MAPE	= mean absolute percentage error, dimensionless, %
R ²	= coefficient of determination, dimensionless
Z-score	= statistical score for outlier detection, dimensionless
MW	= molecular weight, m/n, lb/mol [g/mol]

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